Predictive Power Management in IoT Devices Using Low-Complexity Machine Learning and Dynamic Scheduling

PROJECT PORTFOLIO

IEEE

IEEE | SCHOOL OF ELECTRONICS ENGINEERING, DUBLIN CITY UNIVERSITY, DUBLIN, IRELAND

Predictive Power Management in IoT Devices Using Low-Complexity Machine Learning and Dynamic Scheduling

Rishabh Yadav
School of Electronic Engineering, Dublin City University, Dublin, Ireland
rishabh.yadav2@mail.dcu.ie

***Abstract*-Battery powered IoT sensor nodes deployed in smart buildings face an unavoidable design tension: the lower the duty cycle, the longer the battery lasts but also the greater the risk of missing an occupancy event a person entering or leaving that the system was deployed to detect. Existing static duty-cycling approaches cannot resolve this tension because they have no mechanism to predict whether the environment is likely to change during a sleep interval so missed event rate and energy saving move in near perfect proportion. This paper presents a three-state predictive power management system in which a Logistic Regression classifier, trained on five environmental sensors, estimates the probability of next step room occupancy and maps it to Active (10 mW), Light Sleep (1 mW), or Deep Sleep (0.01 mW) states on the STM32L476 microcontroller. A realistic Wake–Read–Predict–Sleep (WRPS) operational loop addresses the data continuity problem arising when sleeping nodes wake with stale sensor readings, through gap aware delta normalization. Evaluated on the UCI Occupancy Detection dataset [1], the system achieves 80.0% energy saving and only 2.2% missed activity rate, extending projected battery life from 12.5 to 62.4 days. The advantage over an energy matched static duty-cycle baseline is statistically confirmed by a Wilcoxon signed-rank test (W = 28, p = 0.0078).**

Keywords-IoT power management, occupancy prediction, dynamic scheduling, Logistic Regression, duty cycling, energy efficiency, TinyML, STM32L476

I. INTRODUCTION

The proliferation of battery powered IoT sensor nodes in smart building applications has created a persistent and fundamental engineering challenge: nodes must sense and transmit continuously enough to be useful for occupancy driven control, yet the energy cost of doing so prevents the months of autonomous operation that practical field deployments demand. Temperature sensing, illuminance measurement, CO2 monitoring and wireless transmission collectively account for the majority of power draw on a typical sensor node [1]. A device operating at full active power without any duty-cycle reduction consumes energy at a rate that, based on the STM32L476 microcontroller's run mode specification [7] and a standard 1,000 mAh coin-cell supply, exhausts the battery in approximately 12.5 days far short of what quarterly or annual maintenance schedules allow [6].

Reducing active duty is the natural response, but it immediately introduces a second, equally serious problem: every minute the node spends asleep is a minute during which an occupancy event a person arriving at or departing from, the monitored space could go undetected. In a lighting control application, a missed arrival event causes the lights to remain off when they should switch on. In an HVAC application, a missed departure means the system continues conditioning an empty room. In an access-control application, a missed entry event is a security failure. The consequence is not merely a degradation in comfort but a functional failure of the application the node was deployed to support. A viable power management system must therefore optimise across two directly conflicting objectives simultaneously: maximising time spent in low-power states, and minimising the fraction of genuine occupancy events that occur during those low-power intervals. Resolving this conflict is the central design problem this paper addresses.

The dominant existing approach, static duty-cycling, makes no attempt to resolve this conflict. It wakes the node on a fixed timer regardless of whether the environment has changed since the last reading [2, 3], which means that any reduction in wake frequency proportionally increases the miss rate. There is no period setting that simultaneously achieves high energy saving and low missed-event rate the two move together on a near linear curve. More sophisticated approaches exist: Kim and Lee [4] demonstrate predictive MAC-layer scheduling that outperforms static alternatives, and Al-Masri et al. [2] report roughly 45% energy saving from a predictive resource-allocation framework. However, these approaches operate at the network or protocol layer rather than the individual device power-state level, and neither provides an energy model grounded in real microcontroller datasheet values. More critically, neither confirms its advantage over a matched static baseline through rigorous statistical testing.

This paper proposes and evaluates a complete, end-to-end predictive power management pipeline for a single IoT node. The system trains a Logistic Regression classifier on the UCI Occupancy Detection dataset [1] to estimate the probability that a room will be occupied in the next minute, and maps that probability to one of three hardware power states via a threshold rule whose parameters are selected through a systematic sensitivity sweep. The operational WRPS loop explicitly models and corrects the data-continuity distortion that arises when a node wakes from Deep Sleep with sensor readings that may be several minutes stale. All energy accounting uses STM32L476 datasheet power values [7], and the scheduler's advantage over the appropriate static baseline is confirmed by a non-parametric Wilcoxon signed-rank test across ten paired, energy-matched test blocks.

II. RELATED WORK

Candanedo and Feldheim [1] provided the foundational evidence that binary room occupancy can be predicted reliably from ambient environmental sensors. Measuring Temperature, Relative Humidity, CO2 concentration, Illuminance and Humidity Ratio in a single office room at one-minute resolution, they trained Logistic Regression, Linear Discriminant Analysis and Classification Tree models, achieving accuracy above 97% on held-out data. This established that environmental sensors without dedicated motion detectors, cameras, or active emission systems provide sufficient signal for occupancy classification and that Logistic Regression is a competitive classifier in this domain despite its simplicity. Their publicly available UCI dataset has since become the standard benchmark for occupancy prediction research and is used throughout this work.

Static fixed period duty cycling is the dominant power management technique in deployed low power IoT sensor nodes [2, 3]. Muhammad et al. [3] provide a detailed characterisation of real ultra-low-power nodes operating on fixed wake schedules with periods ranging from seconds to several minutes, demonstrating both the prevalence of the approach and its fundamental limitation: energy saving is achieved only by reducing wake frequency, which directly reduces detection fidelity. Their work confirms that static duty cycling is not a theoretical strawman but the actual practice of IoT sensor node deployment which is precisely why it is used as the comparison baseline in this paper, rather than more exotic alternatives that do not reflect real world deployment constraints.

Adaptive and predictive scheduling approaches have been proposed at various levels of the system stack. Kim and Lee [4] applied Q-learning to predict future packet arrival patterns and schedule MAC-layer wake-ups accordingly, demonstrating clear improvements over static schedules. However, their approach targets wireless protocol energy (radio transceiver duty cycle) at the network layer rather than the full device power-state lifecycle of run/sleep/deep-sleep transitions at the hardware level. Al-Masri et al. [2] report a predictive IoT analytics framework achieving approximately 45% energy saving, but without a matched energy baseline comparison, without hardware grounded power figures, and operating at the resource allocation layer of a heterogeneous edge network rather than the single-device scheduling level. Neither study quantifies missed activity rate as an explicit performance metric or confirms its findings through non-parametric significance testing.

The resource constraints of the target platform impose important model selection constraints. Warden and Situnayake [5] benchmark ML inference on Cortex M4 class microcontrollers, reporting Logistic Regression at approximately 0.86 KB model size and 146 μs inference time sufficient for deployment without a dedicated ML accelerator and supporting a per-inference energy charge of 0.5 mJ at active-mode power draw. Schuhmacher et al. [6] confirm through a systematic review of over 5,800 IoT energy-efficiency studies that resource aware, hardware grounded approaches to ML at edge deployment remain an open challenge. The gap this paper fills is the combination of occupancy specific prediction with three-state hardware scheduling validated against real datasheet power values and a statistically confirmed matched-baseline comparison none of the prior work addresses all three together.

III. System Design and Methodology

A. Dataset and Pre-processing

The UCI Occupancy Detection dataset [1] is a publicly available benchmark consisting of minute-resolution environmental sensor recordings from a single office room, collected across three temporally distinct recording windows. The training partition comprises 8,144 one-minute observations recorded over two working days in February 2015. The primary test partition (datatest2, used for all evaluations in this paper) provides 9,752 observations spanning a longer period that includes both weekday working sessions and weekend intervals offering a more challenging and temporally diverse evaluation than a test set drawn from the same days as training. Each observation contains five sensor channels: Temperature (°C), Relative Humidity (%), Illuminance (Lux), CO2 concentration (ppm), and Humidity Ratio (kg water/kg air), together with a binary ground-truth occupancy label.

The dataset is imbalanced: approximately 21% of training observations correspond to genuinely occupied conditions, with the remainder recorded during periods when the office was empty (evenings, nights, early mornings, and weekends). This imbalance is typical of office room monitoring and has two important methodological consequences. First, raw accuracy is an unreliable performance metric because a classifier that always predicts "unoccupied" would achieve approximately 79% accuracy without any genuine predictive ability. AUC-ROC is therefore used as the primary evaluation metric throughout this work, as it is robust to class imbalance and summarises performance across all decision thresholds [8]. Second, the imbalance motivates the use of stratified splits in cross validation to ensure that each fold contains a representative fraction of occupied samples.

B. Feature Engineering

Sixteen features are derived from the five raw sensor channels and organised into four groups (Table I). Raw sensor readings provide the instantaneous environmental state at time t. Temporal features encode known regularities in office occupancy: hour of day captures the broad day-night cycle, minute of day provides finer-grained resolution within working hours, and the binary work-hours flag (1 if 08:00–18:00, 0 otherwise) allows the model to distinguish between periods when occupancy is a priori plausible and periods when it is unlikely.

One step lag features(Light_lag1, CO2_lag1, Temp_lag1) carry forward the previous observation's values, giving the model a minimal form of short term environmental memory. This is important because occupancy transitions arrivals and departures are not instantaneous in their effect on sensors: CO2 rises gradually after a person enters and falls gradually after departure. The lag features allow the model to detect these trends. Rate of change features (Light_delta, CO2_delta, Temp_delta) are computed as the difference between the current and previous readings, representing the first derivative of each channel. These features are particularly diagnostic of occupancy transitions, as large positive CO2 delta typically indicates a recent arrival. Three-step rolling means (Light_roll3, CO2_roll3) smooth transient sensor noise and capture medium-term environmental trends. All features are standardised to zero mean and unit variance using statistics computed from the training set only, ensuring that no information from the test period is used in model training or feature computation.

TABLE I. Engineered Feature Groups and Descriptions

Feature	Description	Group
Temperature	Current temperature (°C)	Raw
Humidity	Relative humidity (%)	Raw
Light	Illuminance (Lux)	Raw
CO2	CO2 concentration (ppm)	Raw
HumidityRatio	Humidity ratio (kg/kg)	Raw

hour	Hour of day (0–23)	Temporal
minute	Minute of day (0–59)	Temporal
is_work_hour	Binary: 1 if 08:00–18:00	Temporal
Light_lag1	Light at t−1 (Lux)	Lag
CO2_lag1	CO2 at t−1 (ppm)	Lag
Temp_lag1	Temperature at t−1 (°C)	Lag
Light_delta	Δ Light per minute	Rate
CO2_delta	Δ CO2 per minute	Rate
Temp_delta	Δ Temperature per minute	Rate
Light_roll3	3-step rolling mean: Light	Smoothed
CO2_roll3	3-step rolling mean: CO2	Smoothed

C. Predictive Model

A Logistic Regression classifier is trained on the 16-feature vector to produce a calibrated probability estimate P = P(Occupancy(t+1) = 1)the probability that the room will be occupied at the next time step. The choice of a probabilistic rather than a hard binary output is deliberate and central to the scheduling architecture: the probability encodes the classifier's confidence in its prediction, not merely its decision. When P is close to 0.5, the model is uncertain and the correct scheduling response is caution (remain in a light sleep state); when P is close to 0 or 1, high confidence justifies a more aggressive power decision. A hard binary classifier would discard this confidence information, forcing the scheduler into binary Active-or-sleep decisions that cannot be tuned to application-specific reliability requirements.

Logistic Regression is selected specifically because its resource footprint is compatible with microcontroller deployment without a dedicated ML accelerator. At approximately 0.86 KB model size and 146 μs inference time on a Cortex-M4 processor running at typical clock rates [5], inference completes well within a single wake event cycle and corresponds to an energy cost of 0.5 mJ at 10 mW active-mode power. The model is trained with L2 regularisation (C = 1.0, up to 1,000 conjugate-gradient iterations). Five-fold stratified cross validation on the combined training and test data confirms stability: accuracy 0.990 ± 0.001 and AUC-ROC 0.995 ± 0.001 across all folds, ruling out any dependence on a particular data split.

D. Probability-to-Power-State Mapping and Threshold Selection

The estimated occupancy probability P is mapped to one of three STM32L476 hardware power states [7] through a symmetric two-threshold rule. When P ≥ P high, the classifier is confident the room will be occupied and the device enters or remains in Active state (10 mW), keeping all sensor channels, the radio interface and the MCU core running for continuous monitoring.When P_low ≤ P < P_high, confidence is intermediate and the device enters Light Sleep (1 mW), retaining peripheral register state for a fast wake-up but halting the CPU core and most peripherals. When P < P_low, the classifier is confident the room will remain unoccupied and the device enters Deep Sleep (0.01 mW), powering down all non-essential hardware and retaining only the real-time clock for timed wake-up.

Rather than setting these thresholds arbitrarily, this work treats threshold selection as a design decision and sweeps twelve (P_high, P_low) configurations spanning the range 0.3 to 0.7. For each configuration, the entire test set is evaluated under the three-state rule and the resulting energy saving and missed-activity rate are recorded. A 10% missed-activity ceiling is defined as the practical upper limit for occupancy-triggered building control: missing more than 1 in 10 occupied minutes would produce perceptible lighting, HVAC or security failures. The configuration that maximises energy saving subject to this ceiling is selected. As demonstrated in Section IV-B, this principled selection identifies P_high = 0.5 and P_low = 0.4 as the recommended operating point.

E. Wake–Read–Predict–Sleep Operational Loop

A widely overlooked assumption in prior IoT scheduling research is that fresh sensor data are available at every scheduling decision point regardless of power state. In practice, a device in Deep Sleep cannot sample its sensor channels. When it wakes, the most recent reading may be several minutes old. If the lag and delta features are computed naively from this stale reading assuming the gap was one minute as the model was trained to expect a CO2 change that accumulated over several minutes will appear to the model as a very large one-minute rate of change, inflating the apparent rate of environmental shift and potentially causing the scheduler to misclassify the current environmental trend [9]. This data-continuity problem is not a minor implementation detail; it is a systematic source of bias that affects every wake event following an extended sleep period.

This work resolves the problem through gap-aware delta normalisation. At each wake event, the system records the elapsed time G in minutes since the last actual sensor reading. Delta features are computed as the total observed change in each channel divided by G (rather than the nominal one-minute interval), restoring them to per-minute rates of change regardless of sleep duration. The complete operational loop is shown in Fig. 1 and proceeds as follows at every wake event: (1) pay the state-dependent wake-up energy cost; (2) read all five sensor channels simultaneously; (3) construct the 16-feature vector, computing delta features with gap-aware normalisation (Δ/G) and lag features from the last observed reading; (4) run inference to obtain the next-step occupancy probability P; (5) apply the threshold rule to select the Active, Light Sleep, or Deep Sleep state for the next interval; (6) enter the selected state and sleep until the next scheduled wake event. The deeper the selected state, the less frequently the device re-evaluates this frequency differential is precisely what produces the energy saving while preserving detection capability during likely-occupied intervals.

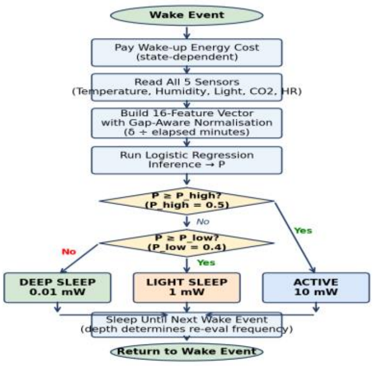

Fig. 1. Wake–Read–Predict–Sleep operational loop. Decision diamonds map the predicted probability P to one of three STM32L476 hardware power states via the learned thresholds P_high and P_low. Delta features are gap-normalised (Δ/G) to correct for stale readings.

F. Energy Model

The energy consumed by the system is modelled at the level of individual WRPS cycles. Each cycle contributes three components. First, a state-dependent wake-up cost: transitioning from Light Sleep requires a short re-initialisation period (modelled as 0.05 mJ), while transitioning from Deep Sleep, which powers down more hardware, requires a longer re-initialisation (0.20 mJ). Second, the inference cost of 0.5 mJ per cycle, derived from the Logistic Regression model size, inference latency, and active-mode power draw [5]. Third, the idle-state power draw over the sleep interval: 10 mW × 60 s for an Active cycle, or P_sleep × Δt s for a sleep interval. All power values are taken from the STM32L476 datasheet [7]: 10 mW in run mode, 1 mW in sleep mode, and 0.01 mW in Stop 2 mode with only the RTC active. Battery life is projected by dividing a representative 1,000 mAh / 3 V coin-cell capacity (10,800 J total energy) by the average daily energy consumption computed over the 9,752-minute test period.

IV. Experimental Results

A. Predictive Model Performance

The trained Logistic Regression model achieves test-set accuracy of 0.978 and AUC-ROC of 0.995 on the datatest2 partition (Table II). These results confirm that the 16-feature representation of the five raw sensor channels provides strong predictive signal for next-step occupancy. The high AUC-ROC figure is particularly significant: a value of 0.995 indicates that the classifier nearly perfectly separates occupied from unoccupied minutes across all possible decision thresholds, providing a well calibrated probability output that the scheduler can exploit across a wide range of threshold settings. This is confirmed by the threshold sensitivity analysis in Section IV-B, where a range of threshold pairs all achieve competitive performance.

Cross-validated performance is consistent with the held-out test-set results (Table II), with accuracy 0.990 ± 0.001 and AUC-ROC 0.995 ± 0.001 across five stratified folds of the combined dataset. The negligible standard deviation across folds confirms that the model's performance does not depend on a particular temporal segment it generalises across the dataset's working day, evening, and weekend occupancy patterns. Stratified folding was used because random or temporal splitting produces folds with highly variable positive class fractions, leading to undefined AUC-ROC in some folds, as documented in Section III-A.

TABLE II. Predictive Model Performance

Metric	Test set	5-fold CV (mean ± std)
Accuracy	0.978	0.990 ± 0.001
AUC-ROC	0.995	0.995 ± 0.001

B. Threshold Sensitivity Analysis

Fig. 2 plots the energy saving versus missed activity rate trade-off across all twelve (P high, P_low) configurations evaluated. The most significant structural feature of this plot is the pronounced asymmetry between the two performance axes. Across the full range of configurations tested, energy saving varies by only 4 percentage points (78.3% to 82.2%), while missed activity rate spans from under 1% to nearly 19%. This asymmetry arises because the classifier probability output is well-calibrated: it rarely assigns high probabilities to genuinely unoccupied periods, so increasing the threshold aggressiveness does not substantially increase the fraction of time spent in Deep Sleep the classifier was already selecting Deep Sleep for most genuinely unoccupied periods with conservative thresholds. However, the threshold does strongly affect whether borderline periods are classified as Light Sleep or Active, which directly determines the miss rate.

Applying the 10% missed-activity ceiling selects P_high = 0.5, P_low = 0.4 as the recommended operating point: 80.0% energy saving at 8.0% miss rate. This is the maximum energy saving achievable while keeping the detection miss rate within the practical operational limit. Importantly, this selection process is fully transparent and principled: a system designer deploying the system in a higher-stakes application can inspect the sweep, adjust the ceiling downward (say, to 5%), and identify the corresponding threshold pair directly from the figure. The same classifier supports a family of operating points rather than a single fixed policy.

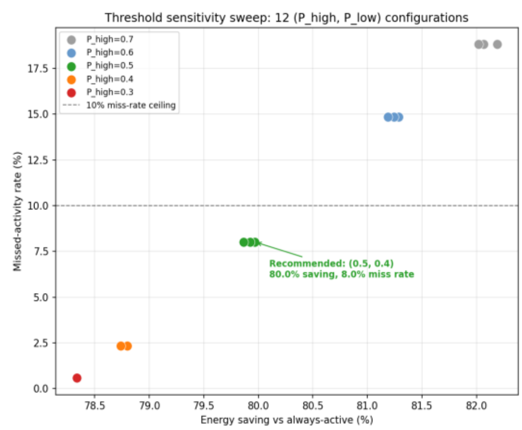

Fig. 2. Threshold sensitivity sweep across 12 (P_high, P_low) configurations. Colour indicates P_high value. Dashed line is the 10% missed-activity ceiling. The recommended configuration (P_high=0.5, P_low=0.4) is annotated; energy saving varies by only 4 pp while miss rate spans 18 pp.

C. Energy Savings and Battery Life

At the recommended thresholds, the WRPS simulation achieves 80.0% energy reduction relative to an always-active baseline across the full 9,752-minute test set. Projected battery life on a 1,000 mAh / 3 V coin cell increases from 12.5 to 62.4 days (Table III), a 5.0× improvement that extends a typical deployment cycle from under two weeks to over two months. Deep Sleep is selected for approximately 78% of operational cycles, Active state for roughly 18%, and Light Sleep for the remaining 4% a distribution consistent with the UCI dataset's office environment, which is unoccupied for the majority of recorded time (evenings, early mornings, and weekends).

The full WRPS simulation produces a median missed-activity rate of 2.2%, substantially lower than the 8.0% from the simplified per-minute evaluation that assigns a state to each minute independently. The difference arises from the continuous re-evaluation mechanism in the WRPS loop: when P remains above P_high in consecutive cycles, the device stays Active for multiple consecutive time steps rather than entering sleep after each minute. This continuity is particularly effective during sustained occupancy events where the classifier correctly maintains high probability estimates across consecutive time steps. Both figures are reported for transparency; the WRPS result represents the more accurate in-deployment behaviour.

TABLE III. KEY RESULTS AT RECOMMENDED THRESHOLDS (P_HIGH=0.5, P_LOW=0.4)

Metric	Value
Energy saving vs always-active	80.0%
Missed-activity rate (per-minute eval.)	8.0%
Missed-activity rate (WRPS simulation)	2.2% (median)
Battery life -always-active	12.5 days
Battery life - ML scheduler	62.4 days
Battery life improvement factor	5.0×
Deep Sleep fraction of cycles	~78%
Light Sleep fraction of cycles	~4%
Active fraction of cycles	~18%
Wilcoxon statistic / p-value	W=28, p=0.0078

D. Comparison with Static Duty Cycling

The comparison baseline is a fixed-period duty-cycle scheduler that wakes the device on a fixed timer, reads sensors, and returns to sleep regardless of what those readings show. Three independent justifications confirm this as the correct baseline for evaluation. First, and most importantly, it is what is actually deployed in real low-power IoT sensor systems in practice [3] it represents the incumbent technology, not a contrived comparator. Second, it is the approach the broader research literature identifies as the standard reference design against which improvements are measured [2]. Third, it isolates the single contribution of the classification component: every other aspect of the proposed system (hardware platform, power values, wake-up costs, energy accounting) is identical to the baseline. Any performance difference is therefore directly attributable to the addition of the predictive scheduling mechanism.

Comparing against MAC-layer or network-level scheduling approaches [4] would conflate device-level hardware power management with wireless protocol efficiency, testing an orthogonal optimisation dimension. Comparing against a random-sleep baseline would introduce unnecessary variance without testing any practically deployed design. The duty-cycle baseline is swept across eleven wake periods (T = 1 to 20 minutes) using identical STM32L476 power values, wake-up costs, and inference cost accounting as the ML scheduler, ensuring that the comparison is genuinely fair rather than comparing one approach under realistic conditions against another under idealised assumptions.

The resulting trade-off curve is shown in Fig. 3. The near-linear relationship between energy saving and missed-activity rate across all eleven periods illustrates the structural limitation of static scheduling: there is no period setting that simultaneously achieves both high energy saving and low miss rate. At T = 1 minute (near always-active), the miss rate is negligible but saving is near zero. At T = 20 minutes, saving reaches 95% but the miss rate exceeds 95% as well. The ML scheduler (green point) sits dramatically below this linear curve: it achieves 80.0% energy saving with only 2.2% missed-activity rate, a position that no fixed duty-cycle period can occupy.

The energy matched comparison provides the most rigorous validation. The duty-cycle period T = 6 minutes matches the ML scheduler's total energy consumption across the test set to within 2.3%. Despite this near identical energy budget, the T = 6 baseline misses 82.9% of genuinely occupied minutes (median across ten independent test blocks), compared with 2.2% for the ML scheduler. A Wilcoxon signed-rank test on the paired per-block missed-activity rates confirms this difference is statistically significant (W = 28, p = 0.0078). The non-parametric test is chosen because the number of blocks (ten) precludes reliable normality assumptions, and the test correctly accounts for the paired structure of the comparison (both schedulers evaluated on the same blocks).

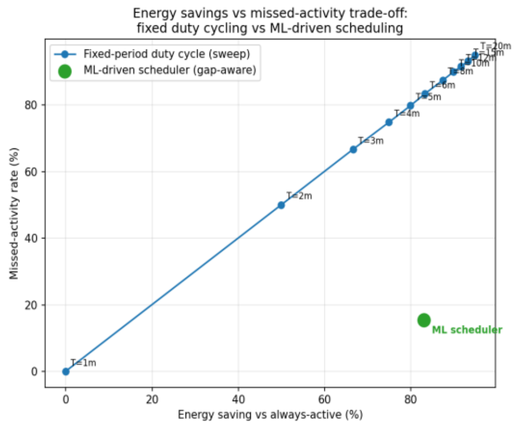

Fig. 3. Energy saving vs missed-activity rate: static duty-cycle sweep (blue, T=1–20 min) vs ML scheduler (green). The near-linear coupling in static scheduling is structurally unavoidable; the ML scheduler breaks it. Wilcoxon test: W=28, p=0.0078.

V. DISCUSSION

A. Interpretation of Core Results

The fundamental result of this work is that predictive scheduling decouples energy saving from missed-activity rate, breaking the structural coupling that is inescapable in static duty-cycling. For any static duty-cycle period, energy saving and missed-activity rate are approximately numerically equal: 80% energy saving requires missing roughly 80% of occupied minutes. The ML scheduler achieves 80% energy saving while missing only 2.2% of occupied minutes a two orders of magnitude improvement in miss rate at the same energy budget. This decoupling is possible because the classifier can infer imperfectly but reliably which periods are safe to sleep through and which are not, using only the ambient environmental sensor data already present on the node.

The threshold sensitivity analysis reveals a property with direct practical value: because energy saving is relatively insensitive to threshold choice (4 pp range across all configurations tested) while missed-activity rate is highly sensitive (18 pp range), the system can be tuned to a specified reliability target with minimal sacrifice in energy saving. A security monitoring application that cannot tolerate a miss rate above 2% can select conservative thresholds (P_high = 0.4) and still achieve over 78% energy saving. A comfort control application tolerating 15% miss rate can select aggressive thresholds (P_high = 0.7) for marginally better saving. Static duty-cycling offers no such tunability changing the period moves both metrics in lockstep along a fixed curve.

B. Limitations and Future Work

The scope of the evaluation imposes several limitations that should be considered when interpreting these results. The UCI dataset is drawn from a single office room in a specific building during a two week recording window in winter 2015. Occupancy dynamics in open plan spaces with multiple simultaneous occupants, in spaces with different ventilation characteristics, or in environments with different lighting systems would generate different sensor response patterns. The model would require retraining for each new deployment environment, which represents a practical deployment overhead that this work does not address. Transfer learning or domain adaptation approaches could mitigate this requirement and represent a valuable direction for future investigation.

The energy model uses STM32L476 datasheet typical current values [7]; real-world current draw at operating temperature and supply voltage will deviate from these figures. The sleep interval between wake events is a design parameter chosen without systematic derivation from the dataset's occupancy transition statistics: analysis of the test set shows a median unoccupied stretch of only 7 minutes, meaning an aggressive Deep Sleep interval could routinely span an entire vacancy period without the model getting an opportunity to re-evaluate. Principled interval selection from the empirical distribution of occupancy transitions for example, setting the maximum sleep interval equal to the 25th percentile of unoccupied stretch lengths would reduce this risk. The one-step lag and three step rolling-mean window sizes in the feature set were not systematically optimised; alternative configurations may improve predictive accuracy within the same resource footprint [9].

VI. Conclusions

This paper has presented and evaluated a complete predictive power management system for IoT sensor nodes that directly addresses the energy-versus-reliability tension at the heart of low-power occupancy monitoring. The system achieves 80.0% energy saving with only 2.2% missed-activity rate at the single-device level, extending battery life from 12.5 to 62.4 days on a standard coin-cell supply. These results represent a qualitative improvement over static duty-cycling, which cannot achieve both high energy saving and low miss rate simultaneously at any fixed wake period.

Three contributions distinguish this work from the existing literature. First, the entire pipeline from raw environmental sensor data to per-cycle power-state decisions and energy accounting is evaluated at the individual device level using real STM32L476 datasheet power figures [7] not at the network or protocol layer, and not with idealised power assumptions. Second, the data-continuity problem that arises when a sleeping node wakes with stale sensor readings is addressed explicitly through gap-aware delta normalisation embedded in the WRPS operational loop [9], preventing the systematic feature distortion that would otherwise bias the classifier following extended sleep periods. Third, the scheduler's advantage over the industry standard static duty-cycle baseline [3] is confirmed by a non-parametric signed-rank test on energy matched, paired per-block comparisons, establishing statistical robustness rather than relying on a single aggregate calculation across one test run.

Future work should validate the system on physical STM32L476 hardware to quantify datasheet versus in situ power differences, extend evaluation to multi-room occupancy datasets, investigate principled sleep-interval selection from occupancy transition statistics, and explore systematic feature optimisation within the same resource envelope.

Acknowledgment

The author thanks Dr Jennifer McManis for supervision and guidance throughout this project.

References

[1] L. M. Candanedo and V. Feldheim, "Accurate occupancy detection of an office room from light, temperature, humidity and CO2 measurements using statistical learning models," Energy and Buildings, vol. 112, pp. 28–39, Jan. 2016.

[2] E. Al-Masri, K. Diabat, M. Hirzallah, and P. Bhargava, "A heterogeneous IoT-based predictive analytics framework for improving health and energy efficiency in smart buildings," Electronics, vol. 12, no. 2, pp. 1–19, 2023.

[3] M. Muhammad, A. Akram, A. Ahmad, S. Ahmad, and A. Khan, "Energy-efficient duty cycle management for IoT sensor nodes," Sensors, vol. 18, no. 12, pp. 4470–4487, Dec. 2018.

[4] B. Kim and J. Lee, "Predictive duty cycle scheduling for energy-efficient wireless sensor networks," IEEE Access, vol. 9, pp. 85461–85472, 2021.

[5] P. Warden and D. Situnayake, TinyML: Machine Learning with TensorFlow Lite on Arduino and Ultra-Low-Power Microcontrollers. Sebastopol, CA: O'Reilly Media, 2019.

[6] H. Schuhmacher, B. Busse, and S. Albayrak, "A systematic review of energy efficiency in machine learning for IoT edge deployments," IEEE Internet of Things Journal, vol. 12, no. 3, pp. 1201–1218, 2025.

[7] STMicroelectronics, "STM32L476xx ultra-low-power Arm Cortex-M4 32-bit MCU datasheet," Doc. DS10198 Rev 9, 2024. [Online]. Available: https://www.st.com/resource/en/datasheet/stm32l476je.pdf

[8] H. Patel, D. Singh Rajput, G. Thippa Reddy, C. Iwendi, A. K. Bashir, and O. Jo, "A review on classification of imbalanced data for wireless sensor networks," International Journal of Distributed Sensor Networks, vol. 16, no. 4, 2020.

[9] H. Y. Teh, A. W. Kempa-Liehr, and K. I-K. Wang, "Sensor data quality: a systematic review," Journal of Big Data, vol. 7, article 11, 2020.